Slope-intercept form

1 State what b and m represent in the equation of a line $y = mx + b$.

2 State the slope and y value of the y-intercept of the following equations.

a $y = -7 - \frac{4x}{3}$

b $y = 9 + 6x$

c $y = -x$

d $y = -x - 8$

3 For each of the following equations:

i Rewrite the equation in slope-intercept form. .

ii State the slope and y value of the y-intercept of the equation.

a $y = 5\,(3x - 2)$

b $y = \frac{16x + 12}{4}$

4 Determine the slope of the following lines.

a $y = \frac{4 - x}{9}$

b $y = \frac{3 - 2x}{8}$

5 Find the equation of the line, in slope-intercept form, which has:

a A slope of -8 and crosses the y-axis at -9

b A slope of -5 and crosses the y-axis at 7

c A slope of $\frac{2}{3}$ and goes through the point $(0, 2)$

d The same slope as $y = 3 - 6x$ and the same y-intercept as $y = -3x - 7$

e Zero slope and goes through the point $\left(0, \frac{2}{13}\right)$

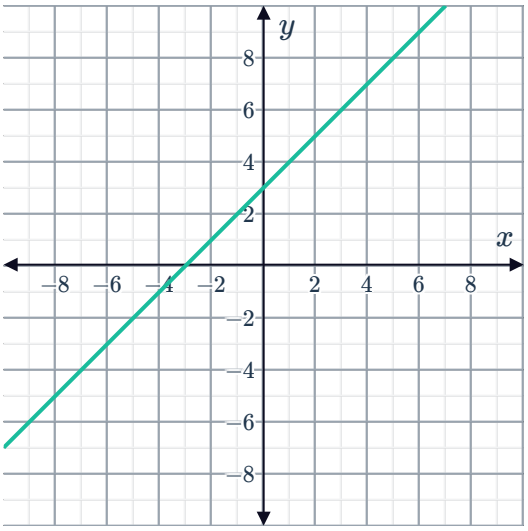
6 A line goes through the points $(0, -1)$ and $(4, 35)$.

a Find the slope of the line.

b Write the equation of the line in the slope-intercept form.

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7 The following is a graph of $y = mx + b$. Sketch the graph of y when b is decreased by 5.



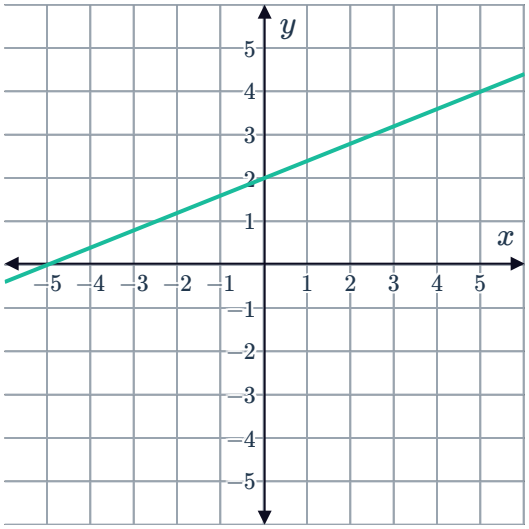
8 Consider the following graphs:

i Find the slope of the line.

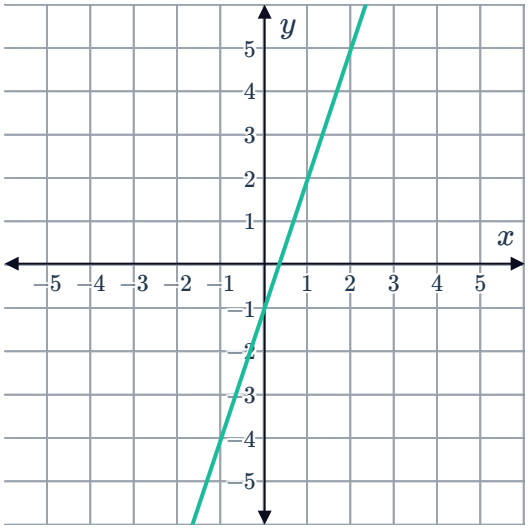
ii Find the y value of the y-intercept of the line.

iii Write the equation of the line in slope-intercept form.

a



b



9 Sketch the graph of the following lines.

a $y = -2x - 1$

b $y = -3x - 3$

10 Given that the line $y = mx + b$ has a slope of -3 and passes through $(-6, -5)$:

a Find the value of b.

b Find the equation of the line in slope-intercept form.

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11 For each of the following:

i Find the value of b.

ii Write the equation of the line in slope-intercept form.

iii Find the x-intercept of the line.

iv Sketch the graph of the line.

a A line has slope $\frac{4}{5}$ and passes through the point $(-10, 4)$.

b A line has slope -2 and passes through the point $(3, -8)$.

12 A straight line has slope -1 and goes through the points $(0, 2)$ and $(a, -6)$.

a Write the equation of the line in the slope-intercept form.

b Find the value of a.

13 For each of the following pairs of equation and table, in which does y have the greater rate of change?

a

- $y = 1 + 8x$
x	0	1	2
y	1	6	11

b

- $y = -3 + 3x$
x	0	1	2
y	-3	2	7

14 Which of the following linear relationships represents a more rapid change in the value of y?

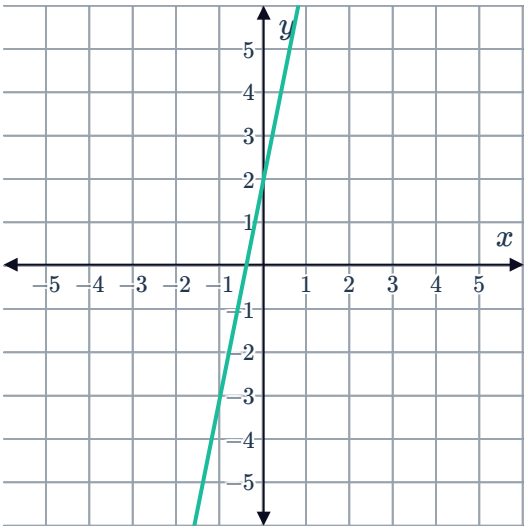
- $y + 6x = 2$
x	0	1	2
y	2	7	12

15 Which of the following linear relationships represents a slower change in the value of y?

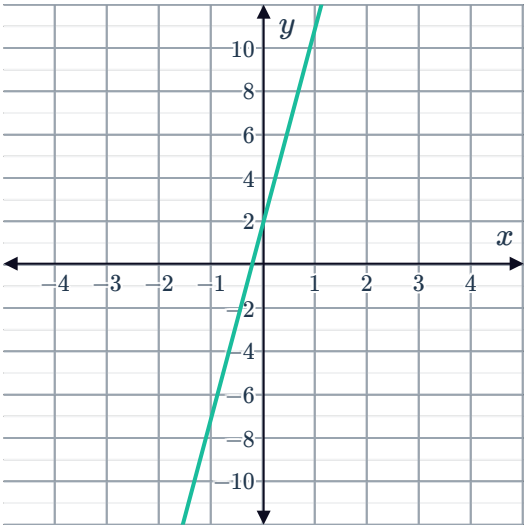
- $y + 7x = 3$
x	0	1	2
y	3	7	11

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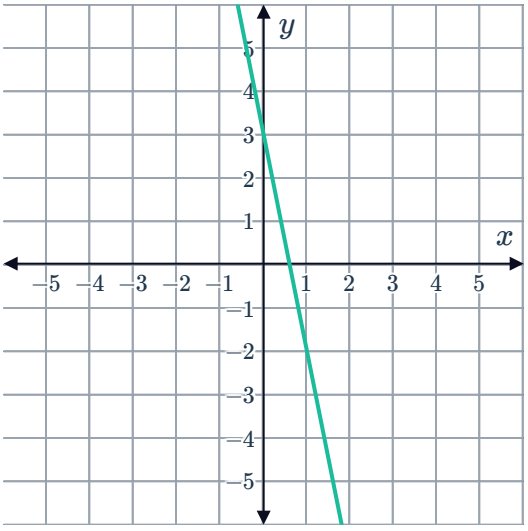
16 Consider the line shown in the following graph.
Does the line $y = 8x + 2$ have a larger or smaller rate of change?



17 Consider the line shown in the following graph.
Does the line $y = 5x + 2$ have a larger or smaller rate of change?



18 Consider the line shown in the following graph.
Are the y-values of the line $y = 8x + 3$ changing more or less rapidly?



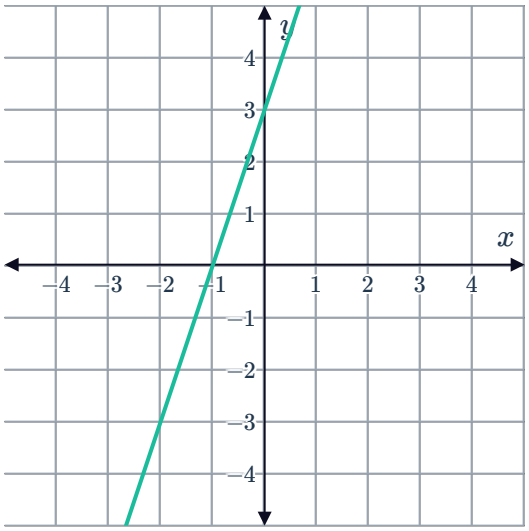
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19 Which of the following linear relationships is the y-value increasing faster?

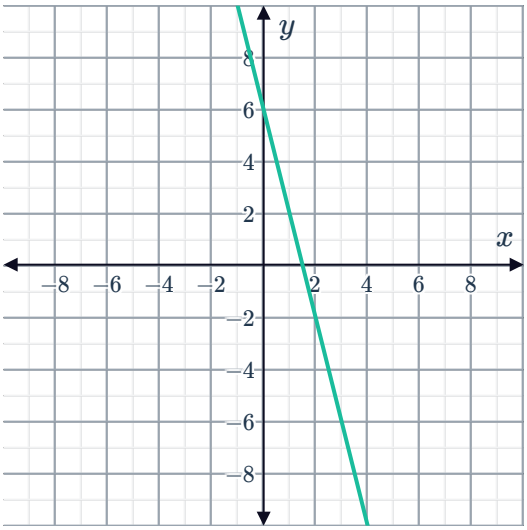
A

x	0	1	2
y	3	7	11

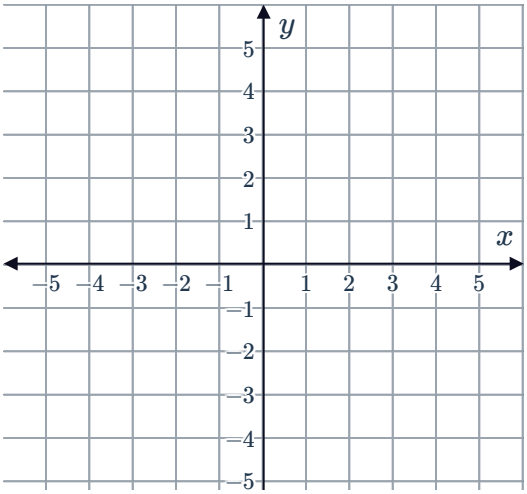
B



20 Consider the line shown in the following graph.
Does the line $y = -4x + 4$ have the larger or smaller y-intercept?



21 Consider the line shown in the following graph.
Does the line $y = -2x + 8$ have a larger or smaller y-intercept?



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22 Some friends decide to go camping for the weekend. They cannot all fit in one car so some of them catch a bus to the campground, which is $450\ \text{km}$ from home.
Those in the car started driving at 8 AM and arrived at the campground at 3:30 PM, driving at a constant speed.
The bus also drives at a constant speed and takes the same route as the car. Its distance y, in kilometers, from home, x hours after leaving, is given by the equation $y = 71x$.

a Determine the speed of the car.

b Determine the speed of the bus.

c Which vehicle was traveling faster?

23 Avril is looking to find the best deal for an internet service to her home, and is comparing two different internet providers.

- Interpute is offering a $\$25$ per month deal where there is no limit to her internet usage.
- Compunet is offering rates according to the amount of internet usage (measured in GB), as shown in the table.

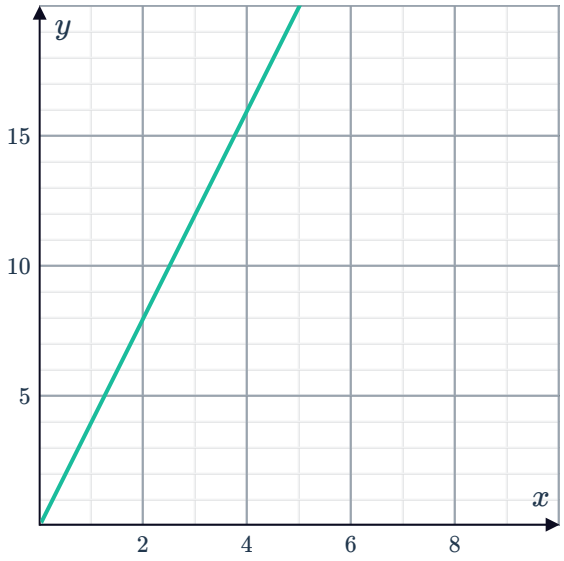
Usage (GB)	Cost
5	$17.50
10	$20.00
20	$25.00
50	$40.00

a If u is the amount of internet usage in a month that would make the two offers cost the same, find u.

b Avril estimates that she will be using about $19\ \text{GB}$ a month.
Which company offers her the better deal?

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24 Mario wants to determine which of two slow-release pain medications is more rapidly absorbed by the body.
For the liquid form, the amount of the medication in the bloodstream is presented in the graph below. The results for the capsule form are presented in the table below.



Time passed (in mins)	Amount in bloodstream (in mgs)
4	24.6
7	42.3
10	60
13	77.7

a At what rate, in milligrams per minute, is the liquid form absorbed?

b At what rate, in milligrams per minute, is the capsule form absorbed?

c In which form is the medication absorbed more rapidly?

Additional questions

25 Consider the following linear equations:

- Line 1: $y = 2 + 6x$
- Line 2: $y = 5x + 6$
- Line 3: $y = 6x$
- Line 4: $y = \frac{x}{6} + 2$
- Line 5: $y = 5 - 6x$
- Line 6: $y = 6x - 2$

State the three lines that have the same slope.

26 For each of the following lines, state whether the following lines have a y-intercept or not. If the line does have a y-intercept, state the coordinates of it.

a $y = 2$ **b** $5x = 4y$ **c** $y = -4x + 1$ **d** $x = 1$

27 Determine whether each of the following lines has the same slope as$y = 6x + 3$ or not.

a $y = 6x$ **b** $y = 3x + 3$ **c** $12x - 2y = 3$ **d** $18x + 3y = 0$

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28 A race car starts the race with 60 gallons of fuel. From there, it uses fuel at a rate of 1 gallon per minute.

a Copy and complete the table of values:

Number of minutes passed (x)	0	5	10	15	20	60
Amount of fuel left in tank (y)						

b Write the equation relating the number of minutes passed (x) and the amount of fuel left in the tank (y).

c Sketch the graph of the function.

29 A cleaner charges an initial fee of $\$110$ plus $\$20$ per hour.
Write an equation to represent the total amount charged by the cleaner, y, as a function of the number of hours worked, x.

30 Neil is running a 30-mile ultramarathon.

a Write an equation to represent the distance Neil has left to run, y, in terms of the number of hours since the start, x.

b The domain has been restricted to $0 \leq x \leq 6$, explain why this restriction has been put in place.



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31 The table shows the water level of a well that is being emptied at a constant rate with a pump where the time is measured in minutes and the water level in feet.

Time (x)	3	7	10
Water level (y)	56.7	48.3	42

a Write an equation in the form $y = mx + b$ to represent the water level (y), as a function of the minutes passed (x).

b Interpret the slope of this linear function.

c Interpret the y-intercept of this linear function.

d Find the water level be after 18 minutes.

e State an appropriate domain constraint on this linear function. Justify your answer.

32 A carpenter charges an initial fee of $\$250$ plus $\$35$ per hour for labor.

a Write an equation to represent the total amount charged, y, by the carpenter as a function of the number of hours worked, x.

b Interpret the slope of this linear function.

c Interpret the y-intercept of this linear function.

d Peta has budgeted $\$2500$ to have a ramp built at the community center to increase accessibilty.
The carpenter works 8 hour days and estimated that the project would take one and a half days. He also priced the materials at $\$1800$.
Determine if Peta has budgeted enough for the ramp. Justify your answer.

33 A diver starts at the surface of the water and begins to descend below the surface at a constant rate. The table shows the depth below the surface of the diver over the first 5 minutes.

Number of minutes passed (x) (min)	0	1	2	3	4
Depth of diver (y) (ft)	0	1.63	3.26	4.89	6.52

a State the rate of change with units of ft/min.

b Write an equation for the relationship between the number of minutes passed (x) and the depth below the surface (y) of the diver.

c Determine the depth of the diver after 6 minutes.

d Find how long the diver takes to reach 11.41 ft beneath the surface.

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e The diver has two air tanks, each of which last for 45 minutes. One is used for going down and one is used for returning to the surface.
Determine an appropriate domain restriction for the function from part (b).

f Assuming the diver comes back up at the same rate they go down, state the equation for the linear function which could describe the return to surface including the domain restriction. Sketch the full scenario as a graph.

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